

ALA-1

Code With Mistake Comments

```
print("Trend Growth Analyzer")
months = int(input("Enter months: "))
i = 0
previous = 0
growth = 0
while i < months:
    value = int(input("Enter monthly value: "))
# Mistake 1: Indentation Error (value should be inside while loop)

    if value > previous:
        growth = growth + 1
    previous = value
    i = i + 1
print("Growth months:", growth)

ratio = growth / months
print("Growth ratio:", ratio)
# Mistake 2: Missing closing bracket ) → Syntax Error

if ratio > 1:
    print("Impossible ratio")
# Mistake 3: Logical mistake (ratio cannot be greater than 1)

if months < 0:
    print("Invalid months")
# Mistake 4: Wrong place for validation (should check before loop)
```

Mistakes Explanation

1. Indentation Error

- `value = int(input(...))` is not inside the `while` loop.
- Python requires proper indentation.
- This causes **IndentationError**.

2. Missing Bracket (Syntax Error)

- `print("Growth ratio:", ratio)` is missing a closing `)`
- Python will show **SyntaxError**.

3. Logical Mistake (Impossible Condition)

- `ratio > 1` condition is not possible.
- Growth cannot be more than total months.
- This condition is logically unnecessary.

4. Validation in Wrong Place

- `months < 0` is checked after the loop.
- Input validation should be done before executing the loop.
- Otherwise program logic becomes incorrect.

Correct Code

```
print("Trend Growth Analyzer")

months = int(input("Enter months: "))

if months <= 0:
    print("Invalid months")
else:
    i = 0
    previous = 0
    growth = 0

    while i < months:
        value = int(input("Enter monthly value: "))

        if value > previous:
            growth = growth + 1

        previous = value
        i = i + 1

    print("Growth months:", growth)

    ratio = growth / months
    print("Growth ratio:", ratio)

    if ratio > 1:
        print("Impossible ratio")
```

OUTPUT:

```
"C:\Users\MALLA ALI MOHAMMAD\AppData\Local\Microsoft\WindowsApps\python3.11.exe" "D:\Collage\Python\Lab Prac\Muffazal-ALA-1.py"  
Trend Growth Analyzer  
Enter months: 2  
Enter monthly value: 200  
Enter monthly value: 300  
Growth months: 2  
Growth ratio: 1.0  
  
Process finished with exit code 0
```